P14 Platformer Slice

CRE132 worksheet

What you'll make

A complete platformer level, 128 tiles long, built entirely from the scripts you've written in earlier samples. There's no new code: one-way planks, a moving platform, crumbling platforms, spikes, patrolling cats you can stomp, coins and gems, checkpoints, lives, a parallax sky, music, a pause menu and a goal.

First you'll take a guided tour to see which sample each piece comes from and how it's set up. Then you'll extend the level yourself, changing one system at a time.



Figure 1: The start of the level: coins, a one-way plank and a gem up high, with the parallax sky behind.

Before you start

1. Unzip `CRE132_P14_PlatformerSlice_Finished.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_P14_PlatformerSlice_Finished` folder. Open it with Unity **6000.3.11f1**. There's no Starter for this sample: you work in the finished level.
2. Open the scene `Assets/_game/Scenes/Platformer/P14_PlatformerSlice`.
3. Press **Play** and play right through. Run (A/D or arrows), jump (Space; hold it for a higher jump), and press **Escape** or **P** to pause. Press Play again to stop.

The route: planks up to a gem, two cats around a raised block, spikes, **checkpoint 1**, a moving platform over a deadly pit, crumbling platforms over another pit, **checkpoint 2**, a high platform with a cat and a row of coins, planks up to a high gem, a spike field crossed on planks, a last cat, and the goal.

1. The player

Select the **Player** and look down its components. Every one is from an earlier sample:

Component	From	What it does here
Better Jump Controller	P02	Coyote time, jump buffering, variable jump height. **Move Speed** 6, **Jump Speed** 15.
Platformer Animator	P06	Idle, run, jump, fall and land animations.
Health	S01, S09	**Max Health** 3, **Destroy On Death** unticked, **On Died** calls **PlayerRespawn > Die**.
Health Tint	S01	Turns the hero redder as health drops.
Player Respawn	P10	Death animation, lose a life, back to the last checkpoint.
Player Hurt	P09	Knockback and a second of invincibility after each hit.

PlayerHurt has one change since P09. When a hit is fatal, it stops before the knockback, so P10's death sequence isn't fighting a knockback for control of the player:

```
        health.TakeDamage(damage);

        // That hit was fatal: leave it to whatever handles death (P10), and don't knock a
        // dead player around.
        if (health.CurrentHealth <= 0)
        {
            return;
        }
```

Look at it: in Play mode, select the Player and watch **Health** and **Player Hurt** while you touch a cat.

2. Enemies and hazards

Each **Cat** (in **Enemies**) has three scripts: P08's **Enemy Patrol** (**Speed** 2, **Ground Layers** Ground) and **Stompable**, and P09's **Hurts Player** (**Damage** 1). The **Spikes** tilemap (in **Grid**) has **Hurts Player** too.

Two scripts could now hurt the player from a cat's side: Stompable and HurtsPlayer. So **Stompable's Damage To Player is set to 0** on every cat, and Stompable has a small change since P08 so that 0 means "don't hurt at all":

```
        else if (damageToPlayer > 0)
        {
            Health playerHealth = collision.gameObject.GetComponent<Health>();
            if (playerHealth != null)
            {
                playerHealth.TakeDamage(damageToPlayer);
            }
        }
```

Stomping still works from Stompable, and side hits come from HurtsPlayer, with knockback.

Look at it: find the cat just before **checkpoint 1**. It's fenced in by a one-tile step, so it can't walk onto the checkpoint and hit you the moment you respawn. Level design matters as much as code.

3. Platforms and pits

All from P04:

- **One-Way Planks** (in **Grid**) is its own tilemap with a **Platform Effector 2D**, so you can jump up through the planks and land on top.
- **Moving Platform** has **Moving Platform** with **Move Offset** X 10, Y 0 and **Speed** 2.5, and a **Kinematic** Rigidbody 2D. It carries you across the first pit.
- The two **Crumbling Platform** objects shake for **Shake Time** 0.5 seconds, fall away, and come back after **Respawn Time** 3. Their sprite is on a child called **Visual**, so it can shake without moving the collider.
- **Fall Death Zone** (P10) is a long trigger under the level with **Fall Death**: falling into a pit takes all your health.

Look at it: select **Moving Platform** in the Scene view. Where does it start, and where does an offset of 10 take it?

4. Pickups, checkpoints and the goal

- The coins and gems (in **Collectables**) are P07's **Coin** and **Gem** prefabs from `Assets/_game/Prefabs/Collectables`: a coin is worth 1 and a gem 5, each with its own sound and sparkle.
- The two **Checkpoint** stones (in **Checkpoints**) have P10's **Checkpoint**.
- The **Goal** has P13's **Level Goal**. The **Level Complete Canvas** has **Level Complete Screen** with **Next Scene** set to `S06_Menus`, so this level's panel offers **Main Menu** and **Replay**.

5. Background, managers and menus

- **Backgrounds** holds P12's three parallax layers, with the same **Follow Amount** values as P12: **Sky (far)** 1, **Far Clouds (middle)** 0.8, **Near Clouds (near)** 0.5.
- The **GameManager** (S02, with P10's lives: **Starting Lives** 3), **Scene Loader** (S04) and **AudioManager** (S05) prefabs. The AudioManager's **Starting Music** is `Placeholder_Chiptune_Loop`.
- **HUD Canvas** (S03) has the **Health Bar**, **Score Panel** and **Lives Panel**.
- **Pause Canvas** (S06) has the pause menu: **Resume**, **Settings** (music and effects volume sliders) and **Main Menu**.

Look at it: this one scene uses more than 25 of your scripts, and every one is something you've written or seen before.

6. Build your own

Change one thing per system, and play after each change. First make your own copy to work in: in the Project window, select `P14_PlatformerSlice`, press **Ctrl+D** and rename the copy `MyPlatformerLevel`. Open it, then go to **File > Build Profiles** and drag it into the **Scene List**. Dying with no lives left and **Replay** both reload the scene by name, which only works for scenes in the list.

1. **More lives:** select **GameManager** in the Hierarchy and set **Starting Lives** to 5. **Test it:** the lives text starts at 5.
2. **A third checkpoint:** select a **Checkpoint**, press **Ctrl+D** and move the copy somewhere between the existing two. Make sure no cat can walk onto it. **Test it:** touch it, die, and come back there.
3. **A second moving platform:** duplicate **Moving Platform** and move it over the crumbling platforms' pit. Try **Move Offset** X 0, Y 4 for a lift, and a different **Speed**. **Test it:** ride it both ways.
4. **A harder cat:** duplicate a **Cat** and set its **Speed** to 4. **Test it:** can you still stomp it?
5. **A different music track:** drag a music file (.ogg or .mp3) into `Assets/_game/Audio/Music`, select the **AudioManager** and put it in **Starting Music**. **Test it:** it plays when the level starts.
6. **A new stretch of level:** open **Window > 2D > Tile Palette** and pick the **Cliffside** palette from the list at the top (it's in `Assets/_game/Tilemaps/Palettes`, with its tiles in `Tilemaps/Tiles/Cliffside`). Pick the **Active Tilemap** in the palette window before painting: **Ground** for ground, **Spikes** for spikes, **One-Way Planks** for planks. Paint a new section past the end, move the **Goal** there, and drop in coins from the prefab. Then make the level big enough for it: select **Camera Bounds**,

click **Edit Collider** on its Polygon Collider 2D and drag its points out to cover the new part (C02's confiner stops the camera at its edge), and make the **Fall Death Zone** wider. If the sky runs out, raise the background sprites' **Width**. **Test it:** the camera follows you all the way, and the pits still kill you.

Check your work

- ☐ You can name the sample that each component on the Player comes from.
- ☐ You know why the cats' Stompable has **Damage To Player** 0.
- ☐ Your copy of the level is in the **Scene List**, and Replay and game over both reload it.
- ☐ Each change you made works, and the rest of the level still plays the same.
- ☐ Compare with the untouched `P14_PlatformerSlice` scene, or a fresh copy of `CRE132_P14_PlatformerSlice_Finished`, if something stops working.

Key points

- **Every system is a component you've seen.** A whole level is many small scripts, each doing one job, combined in the Inspector.
- **Small changes keep scripts working together.** Stompable's 0 damage and PlayerHurt's fatal-hit check are two lines each, and they stop two systems clashing.
- **Level design matters as much as code**: where you put a cat or a checkpoint can make the game fair or unfair.

If something goes wrong

- **Dying with no lives left or pressing Replay gives a "couldn't be loaded" error:** your scene isn't in the **Scene List**.
- **You get hurt twice from a cat's side:** a cat's Stompable **Damage To Player** isn't 0.
- **A new checkpoint does nothing:** it lost its trigger: check **Is Trigger** on its Box Collider 2D.
- **The camera stops before your new section:** the **Camera Bounds** collider doesn't cover it.
- **You fall into a new pit for ever:** the **Fall Death Zone** doesn't reach under it.
- **Painted tiles have no collision:** you painted on the wrong tilemap. Check the **Active Tilemap** in the palette window.

Try it

1. Add a second level after this one: make **Next Scene** on this level's **Level Complete Screen** point at it (P13).
2. Show a "Game Over" panel instead of restarting straight away (P10's Try it).
3. Add a 1-up pickup that calls a new `GameManager.AddLife()`.
4. Design a level of your own from scratch with the same pieces, and swap levels with someone else to play.